Eric Hayden Campbell

Postdoctoral Research Fellow, The University of Texas at Austin

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Research Positions

2024 – present **The University of Texas at Austin**, *Austin, TX*
Postdoctoral Research Fellow, advised by Isil Dillig and Aditya Akella

Summer 2018 **Fujitsu Laboratories of America**, *Sunnyvale, CA*
Research Intern, mentored by Alex Horn

Education

Dec 2024 **Cornell University**, *Ithaca, NY*
Ph.D. in Programming Languages, advised by Nate Foster
Thesis Topic: Verified Configuration of Programmable Networks

May 2017 **Pomona College**, *Claremont, CA*
B.A.s in Computer Science & Mathematics
Thesis: (In)finiteness and Linear Temporal Logic

Awards & Honors

2026 **Distinguished Paper Award**, PLDI, *MatchBox*
2022 **Distinguished Paper Award**, PLDI, *Kleene Algebra Modulo Theories*
2019 **NSF Graduate Research Fellowship**
2018 **Teaching Excellence Award**, Cornell CS, *OOP and Data Structures*, nominated by David Gries

Publications: Formal Foundations of Networking

PLDI '26 **MatchBox: A Semantic Foundation for Data Plane Portability**
Distinguished Paper *Eric Hayden Campbell*, Robert Zhang, Divyanshu Saxena, Aditya Akella, and Isil Dillig

OOPSLA '24 **Computing Precise Control Interface Specifications**
Eric Hayden Campbell, Hossein Hojjat, and Nate Foster

CPP '23 **P4Cub: A Little Language for Big Routers**
Rudy Peterson, *Eric Hayden Campbell*, John Chen, Natalie Isak,
Calvin Shyu, Ryan Doenges, Parisa Ataei, and Nate Foster

POPL '22 Π4: **Dependently-Typed Data Plane Programming**
Matthias Eichholz, *Eric Hayden Campbell*, Matthias Krebs, Nate Foster, and Mira Mezini

NSDI '21 **Avenir: Managing Data Plane Diversity with Control Plane Synthesis**
Eric Hayden Campbell, Bill Hallahan, Priya Srikumar, Carmelo Cascone, Jed Liu,
Vignesh Ramamurthy, Hossein Hojjat, Robert Soulé, Ruzica Piskac, and Nate Foster

ECOOP '19 **How to Avoid Making a Billion-Dollar Mistake: Type-Safe Data Plane Programming with SafeP4**
Matthias Eichholz[*], *Eric Hayden Campbell*[*], Nate Foster, Guido Salvaneschi, and Mira Mezini

Publications: Formal Foundations for Adaptive Operating Systems

PLDI '27 **Refractional Program Logic**
Under Review *Eric Hayden Campbell*, Mark Barbone, Eva Veciana, Neelu Kalani, and Nate Foster

PLDI '27 **Robust Strategy Synthesis for Prediction-based Safety Games**
Under Review *Eric Hayden Campbell*, Aditya Akella, and Isil Dillig

EuroSys '27 **Vulcan: Instance-Specialized, Verifiable Systems Heuristics Through LLM-Driven Search**
Rohit Dwivedula, Divyanshu Saxena, Sujay Yadalam,
Eric Hayden Campbell, Daehyeok Kim, and Aditya Akella

PLDI '26 **Optimal Predicate Pushdown Synthesis**
Robert Zhang, *Eric Hayden Campbell*, Dixin Tang, and Isil Dillig

HotOS '25 **How I Learned to Stop Worrying and Love Learned OS Policies**
Divyanshu Saxena, Jiayi Chen, Sujay Yadalam, Yeonju Ro, Rohit Dwivedula,
Eric Hayden Campbell, Aditya Akella, Christopher J. Rossbach, and Michael Swift

Publications: Logic and Algebra

PLDI '27 **Reading Between the Lines: Finding Hidden Lemmas in Formal Proofs**
Under Review Maxine Xin, *Eric Hayden Campbell*, Sara Ansari, and Isil Dillig

PLDI '22 **Kleene Algebra Modulo Theories**
Distinguished Paper Michael Greenberg, Ryan Beckett, and *Eric Hayden Campbell*

SCiL '21 **Epistemic Semantics in Guarded String Models**
Eric Hayden Campbell and Mats Rooth

TMS '16 **Constructing Integer Matrices with Integer Eigenvalues**
Christopher Towse and *Eric Hayden Campbell*
The Mathematical Scientist, June 2016

Teaching

Computer Science Department, The University of Texas at Austin
Fall 2026 Designer & Instructor, PL Foundations for Systems Researchers, *Informal research seminar*

Academy of Math and Programming, Jane Street Capital
Summer 2023 Advanced Instructor, CS3, *Bespoke course design*
Summer 2022 Instructor, CS3, *Bespoke course design*

Computer Science Department, Cornell University
Spring 2018 TA, OOP and Data Structures, *Teaching Excellence Award*
Fall 2017 TA, OOP and Data Structures

Computer Science Department, Pomona College
Spring 2017 TA, Database Systems
Fall 2016 TA, Programming Languages
Spring 2016 Head TA, Introduction to Computer Science
Fall 2015 TA, Introduction to Computer Science

Service

Reviewing
2026 Program Committee, OOPSLA
2023 Journal Reviewer, Communications of the ACM (CACM)
2022 Journal Reviewer, IEEE/ACM Transactions on Networking (ToN)

Session Previews
2023 Co-Chair, POPL
2023 Session Chair for multiple sessions, POPL
2023 Preview Talk, *Relational & Automated Verification*, POPL

Departmental Service
2024 – 2026 Group Meeting Czar, UToPiA Group, The University of Texas at Austin
2017 – 2024 Colloquium Czar, Cornell University
2018 – 2020 PLDG Czar, Programming Languages Discussion Group, Cornell University

Outreach

2018 – 2022 **4H Career Explorations Conference**
Workshop Designer & Leader, *Networking and Security*

2017 – 2019 **Expand Your Horizons Conference**
Workshop Leader, *Googling With Paper Airplanes*

2015 **Originate, Inc**
Counselor, *Approachable Web Design*

2014 **Tab for a Cause**
Software Engineering Intern, *Website & Browser Extension Front-End*

Talks & Posters

Nov 2026	*Design, Synthesis, and Repair of Verified AI Guardrails*, Symposium Talk, LDOS & InfraAI Symposium
Jun 2026	*MatchBox: A Semantic Foundation for Data Plane Portability*, Conference Talk, PLDI
Jun 2026	*Towards Trustworthy Learning-Driven Systems*, Invited Talk, LDOS Summer School, UW–Madison
Jun 2026	*MatchBox: A Semantic Foundation for Data Plane Portability*, Seminar Talk, MadPL, UW–Madison
Nov 2025	*Performance Contracts*, Poster, LDOS Symposium, UT Austin
Oct 2025	*When P4 Isn't Enough: Specifying the Control Interface*, Recorded Talk, P4 Workshop
Aug 2025	*Relational Verification of Configurable Programs: A Lens-based Approach*, Seminar Talk, Upstate PL
Nov 2024	*Verified Configuration of Programmable Networks*, Dissertation Defense (B-Exam), Cornell
Oct 2024	*Computing Precise Control Interface Specifications*, Conference Talk, OOPSLA
Sep 2024	*Epistemic Semantics in Guarded String Models*, Guest Lecture, Linguistics Semantics Seminar, Cornell
Oct 2023	*Managing Data Plane Diversity with Control Plane Synthesis*, Seminar Talk, EPFL
Jun 2023	*Responsible AI*, Guest Lecture, Hult International Business School
May 2023	*Computing Precise Control Interface Specifications*, Poster, DARPA PRONTO
May 2023	*Managing Data Plane Diversity with Control Plane Synthesis*, Poster, DARPA PRONTO
Jan 2023	*Relational & Automated Verification*, Session Preview Talk, POPL
Oct 2022	*Program Synthesis using Conflict-Driven Learning*, Seminar Talk, PLDG, Cornell
May 2022	*Managing Data Plane Diversity with Control Plane Synthesis*, Project Talk, DARPA PRONTO
May 2022	*Managing Data Plane Diversity with Control Plane Synthesis*, Tech Talk, Intel
Mar 2022	*Managing Data Plane Diversity with Control Plane Synthesis*, Tech Talk, Intel
Jul 2021	*Managing Data Plane Diversity with Control Plane Synthesis*, Invited Talk, CAV P4 Workshop
May 2021	*Managing Data Plane Diversity with Control Plane Synthesis*, Invited Talk, Simons Institute
Apr 2021	*Avenir: Managing Data Plane Diversity with Control Plane Synthesis*, Conference Talk, NSDI
Feb 2021	*Epistemic Semantics in Guarded String Models*, Conference Talk, SCiL
Dec 2020	Thesis Proposal (A-Exam), Cornell
Sep 2020	*Managing Data Plane Diversity with Control Plane Synthesis*, Seminar Talk, Yale University
Apr 2020	*Managing Data Plane Diversity with Control Plane Synthesis*, Invited Talk, P4 Expert Roundtable
Oct 2019	*SafeP4*, Guest Lecture, Network Programming, Cornell
Jul 2019	*Type-Safe Data Plane Programming with SafeP4*, Poster, ECOOP
Apr 2019	*Guarded Impredicative Polymorphism*, Seminar Talk, PLDG, Cornell
Mar 2019	*Decidability of KAT*, Guest Lecture, Introduction to Kleene Algebra, Cornell
Mar 2019	*Definitional Interpreters*, Seminar Talk, Great Works Seminar, Cornell
Feb 2019	*Responsibilities for Correctness*, Guest Lecture, FWS: Ethics, Cornell
Mar 2018	*Functional Pearl: Every Bit Counts*, Seminar Talk, PLDG, Cornell
May 2017	*Temporal NetKAT*, Seminar Talk, PLDG, Cornell
Apr 2017	*(In)finiteness and Linear Temporal Logic*, Thesis Talk, Pomona College